10.8 Applications of First Order Ordinary Differential Equations

The subject of differential equations has vast applications in solving real world problems. The solutions of the differential equations are used to predict the behaviors of the system at a future time, or at an unknown location. In several problems, the rate at which a quantity changes is a given function of the quantity and /or the time. The objective is to find the quantity itself. If x denotes the amount of the quantity present at time t, then the instantaneous rate at which the quantity changes at time t is $\frac{dx}{dt}$. This leads to a differential equation of the form $\frac{dx}{dt} = f(x,t)$. In this section we shall consider this type of problems only. Further, by rate, we mean the instantaneous rate only.

10.8.1 Population growth

Now, we consider the growth of a population (for example, human, an animal, or a bacteria colony) as a function of time t.

Let $x(t)$ be the size of the population at any time t. Although $x(t)$ is integer-valued, we approximate $x(t)$ as a differentiable function and techniques of differential equation can be applied to determine $x(t)$. Assume that population grows at a rate directly proportional to the amount of population present at that time. Then, we obtain

$\frac{dx}{dt} = kx$, where k is the constant of proportionality . … (1)

Here $k > 0$, since the population always increases.

The solution of the differential equation is $x(t) = Ce^{kt}$, where C is a constant of integration. The values of C and k are determined with the help of initial conditions. Thus, the population increases exponentially with time. This law of population growth is called **Malthusian law**.

Example 10.27

The growth of a population is proportional to the number present. If the population of a colony doubles in 50 years, in how many years will the population become triple?

Solution

Let $x(t)$ be the population at time t. Then $\frac{dx}{dt} = kx$.

By separating the variables, we obtain $\frac{dx}{x} = kdt$.

Integrating on both sides, we get, $\log|x| = kt + \log|C|$ or $x = Ce^{kt}$, where C is an arbitrary constant.

Let x_0 be the population when $t = 0$ and obtain $C = x_0$.

Thus, we get $x = x_0 e^{kt}$.

Now $x = 2x_0$, when $t = 50$ and thus, $k = \frac{1}{50}\log 2$.

Hence, $x = x_0 2^{\frac{t}{50}}$ is the population at any time t.

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